Basic of Valuation

Homework Discussion

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Problem 16: Savings Bonds Rates

16a. Exact rate of return if a \$50 bond doubles in 20 years?

Solution 16a

Use the Future Value formula: $FV = PV(1+r)^n$

$100 = 50(1+r)^{20} \implies 2 = (1+r)^{20}$

$r = 2^{1/20} - 1 = 1.03526 - 1 = \mathbf{3.53\%}$

16b. Value in 2029 if \$50 is invested in 2019 at 0.10% per year?

Solution 16b

$PV = 50, r = 0.001, n = 10$ years

$FV = 50(1+0.001)^{10} = 50(1.010045)$

$FV = \$\mathbf{50.50}$

Problem 16: Savings Bonds Rates (c)

16c. If the bond doubles in face value from \$50 to \$100 between 2029 and 2039, what is the rate of return over those last 10 years?

Solution 16c

Note: The starting value in 2029 is \$50.50 (from part b).

$FV = PV(1+r)^n$

$100 = 50.50(1+r)^{10}$

$1.9802 = (1+r)^{10}$

$r = (1.9802)^{1/10} - 1 = 1.0707 - 1 = \mathbf{7.07}\%$

Problem 17: Calculating Present Values

Problem: You want to buy a \$275,000 Ferrari in 9 years. If you can earn 11.2% annually, how much must you invest today?

Solution

Formula: $PV = \frac{FV}{(1+r)^n}$

$PV = \frac{275{,}000}{(1+0.1112)^9}$

$PV = \frac{275{,}000}{2.6241}$

$PV = \mathbf{\$104,797.77}$

Problem 18: Future Value and Timing

Problem: Invest \$5,500 today for 45 years at 10%. What is it worth? What if you wait 10 years to start?

Solution

Scenario 1 (Invest now):
$FV = 5,500(1.10)^{45} = 5,500(72.8905) = \$\mathbf{400,897.66}$

Scenario 2 (Wait 10 years, $n = 35$):
$FV = 5,500(1.10)^{35} = 5,500(28.1024) = \$\mathbf{154,563.39}$

Strategy: Start early! Waiting 10 years cost over \$246,000 in final wealth due to the power of compounding.

Problem 19: Future Value of a Delayed Investment

Problem: Receive \$20,000 in 2 years, then invest it for 6 more years at 6.8%. Total value in 8 years?

Solution

The cash flow occurs at $t = 2$. It compounds for 6 years until $t = 8$.

$FV = PV(1 + r)^n$

$FV = 20,000(1 + 0.068)^6$

$FV = 20,000(1.4859)$

$FV = \$\mathbf{29,718.94}$

Problem 20: Calculating Number of Periods

Problem: Receive \$10,000 in 2 years. Invest at 11% until you have \$60,000. How long will you wait from *now*?

Solution

Step 1: Find time to grow from \$10k to \$60k

$60,000 = 10,000(1.11)^n \implies 6 = (1.11)^n$

$\ln(6) = n\ln(1.11) \implies n = \frac{1.79176}{0.10436} \approx \mathbf{17.17}$ years

Step 2: Total wait time from today

Wait = 2 years (to receive) + 17.17 years (to grow)

Total Wait = **19.17** years

Problem 72: Calculating Interest Rates (IRR)

Problem: 6 payments of \$15,000 (starting today, age 12). At age 18, receive \$32,000/year for 4 years. What is the return?

To find the return, we set the **Present Value (PV) of Outflows** equal to the **PV of Inflows**.

- **Outflows (Investments):** Six payments of \$15,000.
  - First payment today ($t = 0$).
  - This is an **Annuity Due**.
  - Occurs at $t = 0, 1, 2, 3, 4, 5$.
- **Inflows (Benefits):** Four payments of \$32,000.
  - Starts on 18th birthday. Since the child is 12 now, this is in 6 years.
  - This is a **Deferred Annuity**.
  - Occurs at $t = 6, 7, 8, 9$.

Defining the Cash Flows

To find the return, we solve for r where $PV_{Outflows} = PV_{Inflows}$.

Outflows: Savings Phase (Annuity Due)

Six payments of \$15,000 starting **today** ($t = 0$).

$$PV_{out} = 15,000 \times \left[\frac{1-(1+r)^{-6}}{r}\right] \times (1+r)$$

Inflows: College Phase (Deferred Annuity)

Four payments of \$32,000 starting at $t = 6$. Using a 6-year discount factor for the lump sum:

$$PV_{in} = \left(32,000 \times \left[\frac{1-(1+r)^{-4}}{r}\right]\right) \times \frac{1}{(1+r)^5}$$

Trial and Error (Iteration)

We test rates to bracket the Net Present Value (NPV) around zero.

Trial 1: Test $r = 7\%$

- $PV_{out} = 15,000 \times 4.7665 \times 1.07 = \$76,502$
- $PV_{in} = (32,000 \times 3.3872) \times 0.7129 = \$77,281$
- $NPV \approx 77,281 - 76,502 = \mathbf{778}$

Trial 2: Test $r = 8\%$

- $PV_{out} = 15,000 \times 4.6229 \times 1.08 = \$74,891$
- $PV_{in} = (32,000 \times 3.3121) \times 0.6806 = \$72,134$
- $NPV \approx 74,891 - 72,133 = -\$\mathbf{2,757}$

Interpolation for the Final Result

The true rate is between 7% and 8%.

Linear Interpolation

$$r \approx 7\% + \left(\frac{NPV_{7\%}}{NPV_{7\%} - NPV_{8\%}} \right) \times (8\% - 7\%)$$

$$r \approx 7\% + \left(\frac{778}{778 - (-2,757)} \right) \times (8\% - 7\%)$$

Final Calculated Return:

$$7.22\%$$

Problem 73: Indifference Rate (Plan X vs Plan Y)

Problem: Plan X is \$35k perpetuity. Plan Y is \$47k for 15 years. At what discount rate are you indifferent?

Solution

Set the Present Value of both plans equal to each other:

$$PV_X = PV_Y \implies \frac{35,000}{r} = 47,000 \times \left[\frac{1-(1+r)^{-15}}{r}\right]$$

Multiply both sides by r:

$$35,000 = 47,000 \times [1-(1+r)^{-15}]$$

$$\frac{35,000}{47,000} = 1-(1+r)^{-15} \implies 0.74468 = 1-(1+r)^{-15}$$

$$(1+r)^{-15} = 0.25532 \implies 1+r = (0.25532)^{-1/15}$$

$$1+r = 1.0952 \implies r = \mathbf{9.52\%}$$

Problem 74: Perpetuity with Daily Compounding

Problem: \$75k **every other year** forever. First payment at $t = 1$. $r = 7\%$ compounded daily. Value today?

Solution - Step 1: Find Effective 2-Year Rate

First, find the Effective Annual Rate (EAR):
$EAR = (1 + \frac{0.07}{365})^{365} - 1 = 7.2501\%$ Now, find the effective rate for 2 years (r_{2y}): $r_{2y} = (1 + EAR)^2 - 1 = (1.072501)^2 - 1 = \mathbf{15.0258}\%$

Solution - Step 2: Value Today

Because the first payment is at $t = 1$ (not $t = 2$), this is a perpetuity **due** for a 2-year cycle:
$PV = \frac{C}{r_{2y}} \times (1 + EAR) = \frac{75{,}000}{0.150258} \times (1.072501) = \$\mathbf{535,364.66}$

Problem 74: Part 2 (First Payment at Year 4)

Problem: What is the value today if the first payment occurs 4 years from today?

Solution

If payments occur at $t = 4, 6, 8...$, we find the value at $t = 2$ using the perpetuity formula (since $t = 2$ is one "period" before the first payment at $t = 4$):

$$PV_2 = \frac{C}{r_{2y}} = \frac{75,000}{0.150258} = \$499,141.48$$

Now, discount PV_2 back to today ($t = 0$) using the EAR:

$$PV_0 = \frac{PV_2}{(1 + EAR)^2} = \frac{499,141.48}{(1.072501)^2}$$

$$PV_0 = \frac{499,141.48}{1.150258} = \mathbf{\$433,938.74}$$

Problem 75: Ordinary Annuities vs. Annuities Due

The Concept: An annuity due shifts all payments one period earlier (to the beginning of the period) compared to an ordinary annuity.

Future Value (FV) Relationship: In an ordinary annuity, the last payment earns no interest. In an annuity due, every payment earns interest for one additional period:

$$FV_{due} = FV_{ordinary} \times (1 + r)$$

Present Value (PV) Relationship: Similarly, every payment in an annuity due is discounted for one less period than in an ordinary annuity:

$$PV_{due} = PV_{ordinary} \times (1 + r)$$

Comparison Parameters

To clearly distinguish the values, we will use the following real-number example:

- **Periodic Payment (C):** $1,000
- **Interest Rate (r):** 10% (0.10)
- **Duration (n):** 3 Years

Key Distinction

- **Ordinary Annuity:** Payments at the **END** of each year.
- **Annuity Due:** Payments at the **BEGINNING** of each year.

Future Value (FV) Schedule

How much will the account have at the *end* of Year 3?

Ordinary Annuity (End of Year)

Year	Timing	Interest Periods	Calculation	FV
1	End of Yr 1	2	$\$1,000 \times (1.10)^2$	$\$1,210$
2	End of Yr 2	1	$\$1,000 \times (1.10)^1$	$\$1,100$
3	End of Yr 3	0	$\$1,000 \times (1.10)^0$	$\$1,000$
Total				**$3,310**

Annuity Due (Beginning of Year)

Year	Timing	Interest Periods	Calculation	FV
1	Beg of Yr 1	3	$\$1,000 \times (1.10)^3$	$\$1,331$
2	Beg of Yr 2	2	$\$1,000 \times (1.10)^2$	$\$1,210$
3	Beg of Yr 3	1	$\$1,000 \times (1.10)^1$	$\$1,100$
Total				**$3,641**

Present Value (PV) Schedule

What are these future payments worth *today* (Year 0)?

Ordinary Annuity (End of Year)

Year	Timing	Discount Periods	Calculation	PV
1	End of Yr 1	1	$\$1,000/(1.10)^1$	\$909.09
2	End of Yr 2	2	$\$1,000/(1.10)^2$	\$826.45
3	End of Yr 3	3	$\$1,000/(1.10)^3$	\$751.31
Total				**\$2,486.85**

Annuity Due (Beginning of Year)

Year	Timing	Discount Periods	Calculation	PV
1	Beg of Yr 1	0	$\$1,000/(1.10)^0$	\$1,000.00
2	Beg of Yr 2	1	$\$1,000/(1.10)^1$	\$909.09
3	Beg of Yr 3	2	$\$1,000/(1.10)^2$	\$826.45
Total				**\$2,735.54**

The Relationship Revealed

Comparing the two schedules, we can observe the mathematical relationship:

1. **FV Observation:** Each payment in the Annuity Due earns exactly one extra period of interest.
2. **PV Observation:** The first payment in the Annuity Due is not discounted at all because it is received today.

The Scaling Factor

In both cases (FV and PV), the Annuity Due is exactly $(1 + r)$ times larger than the Ordinary Annuity:

- $FV_{Due} = \$3,310 \times 1.10 = \$3,641$
- $PV_{Due} = \$2,486.85 \times 1.10 = \$2,735.54$

Problem 76: Calculating Growing Annuities

Given Variables:

- Target $FV = \$4,500,000$
- Time $t = 40$ years
- Return $r = 9\% = 0.09$
- Salary Growth $g = 3\% = 0.03$
- First Year Salary $S_1 = \$60,000$

Step 1: Solve for required first-year investment (C) Using the FV of a growing annuity formula:

$$FV = C \times \left[\frac{(1+r)^t - (1+g)^t}{r-g}\right]$$

$$4,500,000 = C \times \left[\frac{(1.09)^{40} - (1.03)^{40}}{0.09 - 0.03}\right]$$

$$4,500,000 = C \times 469.313 \implies C \approx \$9,588.50$$

Problem 76: Conclusion

Step 2: Calculate the Savings Rate The user saves a constant percentage (x) of their salary.

$$x = \frac{C}{S_1} = \frac{9,588.50}{60,000}$$

$$x \approx 0.1598 \text{ or } \mathbf{15.98\%}$$

Verification: Since both salary and investment grow at 3%, the ratio remains constant throughout the 40-year period.

Problem 77a: Weekly Interest to APR and EAR

Scenario: 6.3% interest per week.

- **APR (Annual Percentage Rate):** Simple interest calculation (periodic rate $\times$ periods per year).

$$APR = 0.063 \times 52 = 3.276 = \mathbf{327.6}\%$$

- **EAR (Effective Annual Rate):** Accounts for compounding over the 52 weeks.

$$EAR = (1 + r_{periodic})^n - 1$$

$$EAR = (1 + 0.063)^{52} - 1 \approx 23.36 \text{ or } \mathbf{2,336}\%$$

Problem 77b: Discount Interest

Scenario: 6.3% discount interest per week. This means you pay interest upfront. If you "borrow" \$100, you only receive \$100 − \$6.30 = \$93.70.

- **Actual Weekly Rate:**

$$r_{week} = \frac{\text{Interest Paid}}{\text{Net Proceeds}} = \frac{6.30}{93.70} \approx 0.067236 \ (6.72\%)$$

- **APR:** $0.067236 \times 52 = 3.4963 = \mathbf{349.63\%}$
- **EAR:** $(1.067236)^{52} - 1 \approx 28.16 = \mathbf{2,816.3\%}$

Problem 77c: Installment Loan Analysis

Scenario: Net proceeds = \$72.32. Repayment = \$25/week for 4 weeks. This is an annuity problem where we solve for r:

$$PV = PMT \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

$$72.32 = 25 \times \left[\frac{1-(1+r)^{-4}}{r}\right]$$

By trial and error or financial calculator:

- **Weekly Rate (r):** $\approx 14.735\%$
- **APR:** $0.14735 \times 52 = \mathbf{766.22\%}$
- **EAR:** $(1.14735)^{52} - 1 \approx \mathbf{129,545.6\%}$

Trial and Error Setup

From Problem 77c, we have an installment loan with the following terms:

- **Net Proceeds (PV):** \$72.32
- **Weekly Payment (PMT):** \$25.00
- **Number of Weeks (n):** 4

We use the Present Value of an Ordinary Annuity formula:

$$72.32 = 25 \times \left[\frac{1-(1+r)^{-4}}{r}\right]$$

Dividing both sides by 25, we look for a factor ($PVIFA_{r,4}$) equal to:

$$\frac{72.32}{25} = 2.8928$$

Step 1: Trial and Error (Finding the Range)

We test two interest rates to "bracket" the target factor of **2.8928**.

- **Try 14%:**

$$PVIFA_{14\%,4} = \frac{1-(1.14)^{-4}}{0.14} = 2.9137$$

(This is slightly too high, meaning our rate is higher than 14%)

- **Try 15%:**

$$PVIFA_{15\%,4} = \frac{1-(1.15)^{-4}}{0.15} = 2.8550$$

(This is slightly too low, meaning our rate is between 14% and 15%)

Step 2: Linear Interpolation

We assume the change between 14% and 15% is linear. We set up a ratio:

$$\frac{r - 0.14}{0.15 - 0.14} = \frac{2.8928 - 2.9137}{2.8550 - 2.9137}$$

Solving for r:

$$\frac{r - 0.14}{0.01} = \frac{-0.0209}{-0.0587}$$

$$\frac{r - 0.14}{0.01} = 0.3560$$

$$r = 0.14 + (0.3560 \times 0.01) = 0.14356 \text{ or } \mathbf{14.36\%}$$

Note: Using a financial calculator or Excel gives a more precise 14.735%. Interpolation is an approximation.

Step 3: APR and EAR

Using our interpolated weekly rate ($r \approx 14.735\%$):

APR (Nominal Rate)

$$APR = 14.735\% \times 52 = \mathbf{766.22\%}$$

EAR (Effective Rate)

$$EAR = (1 + 0.14735)^{52} - 1$$

$$EAR = (1,402.19) - 1 = \mathbf{140,119\%}$$

Conclusion: Small weekly fees lead to astronomical annual rates!

Problem 78: Present Value of a Growing Perpetuity

The Question: What is the PV if the first payment is C and payments grow by C each period?

Mathematical Setup: This is a specific case where the growth rate g is essentially tied to the initial cash flow. The standard formula for a growing perpetuity (where $g < r$) is:

$$PV = \frac{C}{r - g}$$

Specific Constraint: If the growth *increment* is equal to the initial payment C, then $C_2 = 2C$, $C_3 = 3C$, and so on. This represents a **linear gradient series**. The formula for a perpetual gradient is:

$$PV = \frac{C}{r} + \frac{C}{r^2}$$

(Assuming g in the traditional sense refers to a constant percentage, but the prompt implies an arithmetic growth of C units per period).

Visualizing the Payments

A linear perpetuity with payments $C, 2C, 3C, \ldots$ can be broken into "layers":

- **Layer 1:** A level perpetuity of C starting at $t = 1$.
- **Layer 2:** A level perpetuity of C starting at $t = 2$.
- **Layer 3:** A level perpetuity of C starting at $t = 3$.
- ...and so on, infinitely.

Year:	1	2	3	4	...
Layer 1:	C	C	C	C	...
Layer 2:		C	C	C	...
Layer 3:			C	C	...
Total:	C	$2C$	$3C$	$4C$	...

Valuing the Layers

We know the Present Value of a level perpetuity is $PV = \frac{C}{r}$.

Now, let's find the value of each layer **at the time it starts**:

- **PV of Layer 1 (at $t = 0$):** $\frac{C}{r}$
- **PV of Layer 2 (at $t = 1$):** $\frac{C}{r}$
- **PV of Layer 3 (at $t = 2$):** $\frac{C}{r}$

To find the total value today ($t = 0$), we must discount each of these "PVs" back to the present.

The Total Sum

The total Present Value is the sum of the discounted values of all layers:

$$PV_{Total} = \frac{C/r}{(1+r)^0} + \frac{C/r}{(1+r)^1} + \frac{C/r}{(1+r)^2} + \dots$$

Factor out the constant $\frac{C}{r}$:

$$PV_{Total} = \frac{C}{r}\left[1 + \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \dots\right]$$

The term in the brackets is simply the formula for an **Annuity Due perpetuity** factor, which equals $\frac{1+r}{r}$.

Final Result

Substitute the factor back in:

$$PV_{Total} = \frac{C}{r} \times \left(\frac{1+r}{r}\right)$$

$$PV_{Total} = \frac{C(1+r)}{r^2} = \frac{Cr + C}{r^2}$$

Split the fraction:

$$\mathbf{PV} = \frac{\mathbf{C}}{\mathbf{r}} + \frac{\mathbf{C}}{\mathbf{r^2}}$$

Intuition

$\frac{C}{r}$ is the value of the base level of income.
$\frac{C}{r^2}$ is the present value of all future permanent raises.